Resolution Geometry

The Quantum Bridge: From Quantum Observation Laws to Resolved Geometry, Link by Link

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Abstract

Certified endpoints do not certify a bridge. We build the passage from quantum observations to resolved geometry as a composition of links, each with its own RG status:

quantum observation law → quantum-source descent → coupling declaration → geometric route → geometry observa

The *Composition Theorem* states that the composite is LOCKED only if every link is. A coupling that is declared only up to a class makes the composite a CORRESPONDENCE, even when both endpoints are certified.

Exact and numerical checks then establish each link:

- a field mode's renormalized energy density depends only on second moments, and three homodyne variances are the minimal panel, so $|1\rangle$ and a thermal state are different sources of the same $\langle T \rangle$;
- quantum sources violate pointwise energy conditions (squeezed vacuum), so energy conditions belong to the coupling layer;
- one measured Casimir component plus declared symmetries locks the full Casimir tensor;
- global quantum stress is fixed by one context, while local stress needs a declared ordering;
- geometry walks forward exactly (quantum density, then potential, then linearized Einstein tensor, then the density again), and walking back from outside meets a structural wall at the quantum width;
- semiclassical and outcome-conditioned couplings agree on every linear observable and differ only in fluctuations, so the coupling is FORCED for first moments and CONDITIONAL beyond;
- quantum versus classical coupling is the bridge's open plank, and the entanglement test that would lock it is computed (concurrence 0.156 against 0 for any classical channel).

The bridge is therefore LOCKED (conditional) for the mean geometry of sources identified by second moments, CONDITIONAL for fluctuation-sensitive geometry, and a CORRESPONDENCE for the quantum nature of the coupling. It is not closed, and it is now mapped plank by plank. Ten checks pass on five seeds, and three canaries are caught.

Contents

1 What a bridge must certify **2**

2 Links 1–2: quantum observation and source descent **2**

3 Link 3: coupling as a declared class **3**

4 Links 4–5: geometry walks the bridge **4**

5 The bridge status 4

6 What this does and does not show 4

7 Programme 5

A Reproducibility 5

1 What a bridge must certify

A shared language and two certified endpoints are not a bridge. Three requirements are adopted from a hostile review, in RG terms:

1. *quantum-source descent*: which quantum observations identify which stress target, with unresolved alternatives retained;
2. *coupling closure*: how the source affects geometry, declared as a class with a robustness certificate, not assumed;
3. *composition and stability*: the links agree on shared observables, with controlled errors and derivative regularity.

Theorem 1.1 (Composition)**.** *Let $\mathcal{O} \to T$ be source descent and c a coupling map from sources to geometry. If $\mathcal{K}_\mathcal{O} \subseteq \mathcal{K}_T$ and c is a fixed map, then $\mathcal{K}_\mathcal{O} \subseteq \mathcal{K}_{c\circ T}$, and the composite is identified. If c is declared only up to a class $\mathcal{C}$ whose members differ on the source values, the composite is the correspondence $\{c(T) : c \in \mathcal{C}\}$. In that case it is FORCED only where every member of $\mathcal{C}$ agrees within the declared resolution.*

Proof. Kernel inclusion composes along maps. The class statement is the non-circularity certificate (ladder L57) applied to the coupling class. The finite check is exhaustive over 235 systems with $\mathcal{K}_\mathcal{O} \subseteq \mathcal{K}_T$; an unidentified coupling gives a correspondence (QB05). □

2 Links 1–2: quantum observation and source descent

Theorem 2.1 (Second-moment descent; QB01)**.** *For a single field mode, the renormalized energy density at phase θ is*

$$\langle :X_\theta^2: \rangle = \langle a^\dagger a \rangle + \mathrm{Re}\big(\langle a^2 \rangle e^{-2i\theta}\big).$$

Three homodyne variances (at $\theta = 0$, $\pi/4$ and $\pi/2$) reconstruct it at every phase; the error on random states is below 10^{-10}. Every moment above the second lies in the kernel of the source map.

The states $|1\rangle$ and the thermal state with $\bar{n} = 1$ give identical $\langle T \rangle$, yet their trace distance is 0.75. They differ in stress *fluctuations*: $\mathrm{Var}\, n$ is 0 and 2. A fluctuation-sensitive gravitational task therefore needs fourth moments and a larger panel.

Proposition 2.2 (Casimir: one measured component locks the tensor; QB03)**.** *The force between ideal plates measures $T^{zz} = -\pi^2/(240a^4)$ (with $\hbar = c = 1$). Invariance under boosts parallel to the plates forces $T = A\eta$ on the (t, x, y) block, and conformal tracelessness fixes A. Together they lock*

$$T^{\mu\nu} = \frac{\pi^2}{720a^4}\,\mathrm{diag}(-1, 1, 1, -3),$$

whose energy density is negative. Without tracelessness, A is free, so the declaration is doing work and is named. Casimir-force measurements agree with theory at the level of 15%, 5% and 1% [5, 3, 4].

Proposition 2.3 (Contextual stress; QB04)**.** *Position and momentum are incompatible contexts, with no joint law (Fine; ladder L43). The* global *kinetic stress* $\langle p^2\rangle/2m$ *is identified by the momentum context alone. The* local *kinetic energy density needs a declared operator ordering. Two standard densities,* $\frac{\hbar^2}{2m}|\psi'|^2$ *and* $-\frac{\hbar^2}{2m}\mathrm{Re}(\psi^*\psi'')$*, differ exactly by* $\frac{\hbar^2}{4m}\rho''$*, and both integrate to* $\hbar^2/8m\sigma^2$*. Local quantum stress is a CORRESPONDENCE; global stress is LOCKED.*

No-Go 2.4 (Quantum sources and energy conditions; QB02). Squeezed vacuum ($r = 0.6$) has renormalized energy density as low as $\sinh^2 r - \sinh r \cosh r = -0.349$ at some phases, while its average over a period is $\sinh^2 r \geq 0$. A coherent state never goes negative. The energy conditions assumed by geometric theorems (singularity and positive-mass results) are therefore declarations of the coupling layer. For quantum sources they hold only in averaged forms [6].

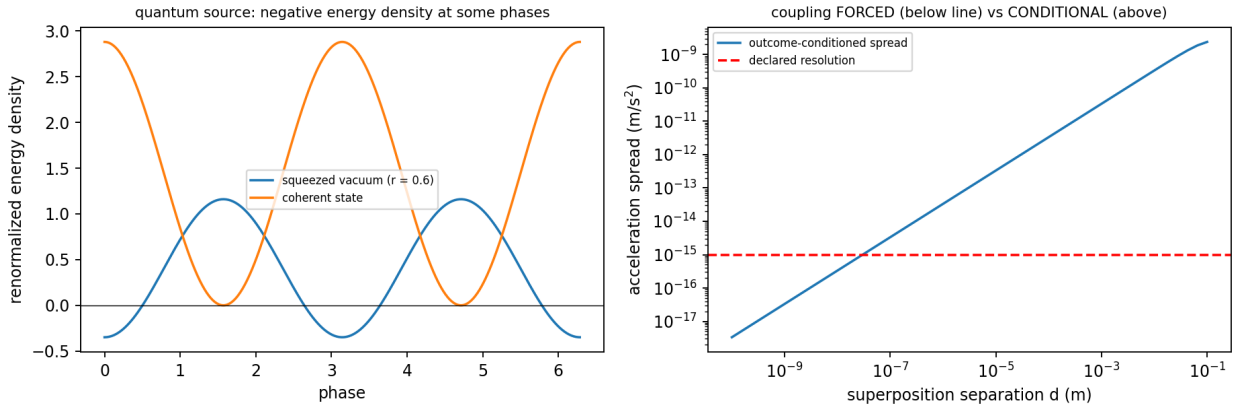


Figure 1: Left: a quantum source with negative energy density at some phases (squeezed vacuum), against a coherent state. Right: for a mass in superposition, the outcome-conditioned acceleration spread against the declared resolution. The coupling class is FORCED below the line and CONDITIONAL above it.

3 Link 3: coupling as a declared class

Theorem 3.1 (Coupling robustness; QB07)**.** *Consider a mass in a superposition of two positions separated by* d*. The semiclassical (mean-field) coupling [7, 8] and the outcome-conditioned coupling predict the* same *mean value of every observable linear in the geometry, by linearity of the field equation in the source at weak field (difference exactly* 0*). They differ only in fluctuations, with spread* $\approx Gmd/R^3$*. At a declared acceleration resolution* τ*, the coupling class is FORCED where the spread is below* τ *and CONDITIONAL above it. For* $m = 1\,kg$ *at* $R = 10\,cm$ *and* $\tau = 10^{-15}\,m\,s^{-2}$*:*

- $d = 10^{-1}\,m$ *and* $10^{-3}\,m$*: CONDITIONAL;*
- $d = 10^{-6}\,m$ *and* $10^{-9}\,m$*: FORCED.*

The macroscopic case is the regime Page and Geilker tested; the gravitational field they observed followed the realized outcome [9].

Proposition 3.2 (The open plank: quantum versus classical coupling; QB08)**.** *Two masses* $m = 10^{-14}\,kg$*, each in a superposition of separation* $250\,\mu m$*, at closest distance* $200\,\mu m$ *for* $2.5\,s$*, acquire a gravitational phase difference* $\Delta\phi = -0.314\,rad$*. The resulting concurrence is* $|\sin(\Delta\phi/2)| = 0.156$*. Any classical coupling, acting by local operations and classical communication, gives* 0 *[10, 11]. Until such an experiment is performed, the quantum nature of the coupling is a CORRESPONDENCE. The experiment that would lock it is specified.*

4 Links 4–5: geometry walks the bridge

Theorem 4.1 (Forward walk is exact; QB06)**.** *For the weak-field metric* $-(1+2\epsilon\Phi)dt^2 + (1-2\epsilon\Phi)d\mathbf{x}^2$*, the Einstein tensor satisfies* $G^t{}_t = -2\epsilon\nabla^2\Phi$ *to first order (computed exactly). A ground-state density* $\rho = m(2\pi\sigma^2)^{-3/2}e^{-r^2/2\sigma^2}$ *gives* $\Phi = -m\,\mathrm{erf}(r/\sqrt{2}\sigma)/r$ *with* $\nabla^2\Phi = 4\pi\rho$ *exactly, and the geometric route returns* ρ *exactly. With* σ *estimated from* 20,000 *simulated position samples* ($\hat{\sigma} = 1.0028$)*, the potential is predicted to* 0.2%*.*

No-Go 4.2 (Walking back meets a wall; QB06, QB09). The exterior field identifies m only (Newton's shell theorem). The quantum width enters as $e^{-r^2/2\sigma^2}$, with relative sensitivity 1.5×10^{-5} at 5σ. It is a STRUCTURAL WALL[τ] unless $\tau < 10^{-5}$. Likewise, the gravity of the Casimir energy (plates of $1\,\mathrm{cm}^2$, gap $1\,\mu$m, mass equivalent $4.8 \times 10^{-31}\,\mathrm{kg}$) is $3.2 \times 10^{-37}\,\mathrm{m\,s^{-2}}$ at 1 cm, 21.5 orders of magnitude below a resolution of 10^{-15}. Soundness is not observability.

5 The bridge status

link	status	basis
quantum observation law	LOCKED (product law over contexts)	L43; Fine
quantum-source descent	LOCKED for second moments; CORRESPONDENCE for local ordering	QB01, QB03, QB04
coupling	FORCED for first moments; CONDITIONAL for fluctuations; CORRESPONDENCE for quantum versus classical	QB07, QB08
geometric route	LOCKED (conditional on the Einstein equation and C^2)	QB06; *The Stress Tensor*
geometry back to source	m LOCKED; width STRUCTURAL WALL	QB06, QB09

By the Composition Theorem, the bridge takes the status of its weakest link:

> LOCKED (conditional) for the mean geometry of sources identified by second moments;
> CONDITIONAL for fluctuation-sensitive geometry (on the coupling class);
> CORRESPONDENCE for the quantum nature of the coupling, until an entanglement experiment is performed.

6 What this does and does not show

Shown.

- The passage from quantum observations to resolved geometry can be built and walked link by link, with exact computations.
- Each link has a typed RG status.
- The bridge's open plank is located, and the experiment that would close it is specified.

Not shown.

- No theory of quantum gravity is proposed.
- Semiclassical coupling is a declared member of a class, not derived.
- Renormalized $\langle T \rangle$ (normal ordering, the Casimir subtraction) is imported from quantum field theory.

- The examples are exact model computations and simulated quantum statistics, not new experimental data.
- "Fully connected" is not claimed: the bridge is conditional, and its weakest plank is open.

7 Programme

1. *Fluctuation-level bridge.* Stress–stress correlators (fourth moments) as an RG task, together with the geometry observables that resolve them.
2. *Ordering correspondence.* Classify local quantum stress definitions, and the observables that would separate them (for example, gravitational response at the scale of $\hbar^2/m$ gradients).
3. *Entanglement plank.* Turn the gravitational-entanglement proposal into a registered RG test, with power analysis for decoherence and Casimir backgrounds.
4. *Curved-background descent.* Quantum-source descent in a resolved curved background (L104), where renormalization depends on the geometry.

A Reproducibility

`rg_qm_bridge_attack.py` (QB01–QB10) passes 10/10 on seeds 20261026, 1, 42, 777 and 2026. Its canaries are:

- claiming that energy identifies the state;
- claiming the two couplings are identical in fluctuations;
- locking the whole bridge.

All are caught. Two corrections were made before registration:

1. A periodic average was first computed on a closed grid, which double-counts the endpoint (bias 1.05×10^{-3}).
2. The rounding helper rounded to decimal places, which erased tiny physical magnitudes (the Casimir energy printed as 0). It now keeps significant figures.

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